

DINITH SHEHAN JIFFRY

C++ Software Engineer | Financial Infrastructure & Exchange Systems

dsjiffry@gmail.com

LinkedIn: <https://www.linkedin.com/in/dsjiffry>

PROFESSIONAL SUMMARY

Software Engineer with 5+ years at the London Stock Exchange Group (LSEG) building back-end financial infrastructure and clearing systems. Specialized in C++11, Linux system programming, Apache Kafka integration, and low-level debugging with GDB and Valgrind. Experienced in leading sprint ceremonies as a Scrum Master and automating developer tooling to reduce manual environment setup.

TECHNICAL SKILLS

- **Languages:** C++ (C++11), Python, Java, SQL (PostgreSQL, Oracle).
- **Systems & Infrastructure:** Multi-threading, Memory Management, Distributed Systems, Linux/Unix, Apache Kafka.
- **Tools & Debugging:** GDB, Valgrind, Google Benchmark, Coverity, Git, Bash.
- **Domain:** Central Counterparty (CCP) Clearing, Market Infrastructure, Automated Testing, Agile/Scrum.

PROFESSIONAL EXPERIENCE

London Stock Exchange Group | Senior Software Engineer (CCP Team) | Mar 2024 – Present

- **A5X Clearing Platform:** Wrote core C++ components for the A5X project, applying code-level logic and data structure optimizations to handle high trade volumes reliably.
- **Kafka Messaging Library:** Designed and implemented a C++ messaging library using Apache Kafka for distributed system I/O, incorporating TLS encryption for client data security.
- **Debugging & Profiling:** Used GDB and Valgrind to locate memory leaks, trace edge-case bugs, and fix resource bottlenecks in high-concurrency environments.
- **Static Analysis:** Managed the Coverity workflow for the team, resolving flagged issues and maintaining release compliance.
- **Code Governance & Compliance:** Owned the Coverity static analysis workflow for the A5X project, to guarantee compliance prior to software releases.
- **Extensible Framework Release Automation:** Built an AI agent to handle routine sprint upgrades of our extensible core framework across client codebases, eliminating manual dependency updates and reducing developer toil.
- **Developer Environment Automation:** Created a set of Bash scripts and Kafka testing tools that automated local topic, partition, and mock server creation for QA and dev teams.
- **Database Tooling:** Built a 255-line dynamic SQL procedure to automate system gateway setup, replacing manual configuration steps.

London Stock Exchange Group | Software Engineer (CCP Team) | Oct 2021 – Mar 2024

- **Qatar Exchange Migration:** Developed C++ back-end logic and stock trading features for the Qatar market migration, matching functional specs for clearing workflows.
- **European Clearing Support:** Maintained software for the RCP/RCL (Paris and London CCP) clearing services, handling bug fixes and operational updates.
- **Testing Suites:** Wrote unit, integration, and regression test suites in C++ to cover edge cases under high load.
- **Scrum Master:** Served as Scrum Master for the CCP development team for one year, running daily stand-ups, planning sessions, and retrospectives.

London Stock Exchange Group | Software Engineer Intern | Feb 2019 – July 2019

- **Benchmarking:** Profiled STL container performance using Google Benchmark to inform container choices in the core engine.
- **Code Maintenance:** Cleaned up legacy C++ code bases to clear compiler warnings and improve readability.
- **SQL Migration:** Validated and migrated datasets from Oracle SQL to PostgreSQL during back-end upgrades.

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT) | Jan 2017 – August 2021

- **BSc (Hons) in Information Technology (Software Engineering)**
- **GPA: 3.29/4.00**
- **Coursework:** Data Structures & Algorithms, Distributed Systems, Parallel Computing, OS Administration, Secure Software Development.

SYSTEMS PROJECTS

- **Turtlebot Maze Navigator (C++/ROS):** Designed an algorithm for autonomous navigation using the ROS framework to solve complex path-finding.
- **Distributed Fire Monitoring (Java/API):** Built a multi-instance safety system designed for independent service recovery and high availability.
- **Security Access Control (C++/Java):** Implemented an access control system using the Apache Felix framework for modular resource management.

CERTIFICATIONS & ACHIEVEMENTS

- **IELTS Academic (2021): Overall Band Score: 9.0** (Listening: 9.0, Reading: 8.0, Speaking: 8.0, Writing: 7.0).
- **Professional Development:** Certificates in Public Speaking and Personal/Professional Development (American College in Higher Education).